The Effect of Verb Type on the Real-time Processing of English *Tough* Constructions by Korean Adult Learners of L2 English

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L2 processing is often characterized as relying more on lexical–semantic cues than on abstract syntactic structures and on interpreting the subject as the agent [1, 2]. Such reliance makes it difficult for learners to process complex constructions in which the theme occupies the subject position. In the real-time processing of the English *tough* construction (TC; *Elmo is easy to see*), this predicts the following L2 processing and interpretation patterns: Learners may initially assign an agentive interpretation to the subject and maintain their initial parse, resulting in an incorrect interpretation; alternatively, they may begin with a subject-as-agent parse but later revise it, thereby arriving at the correct interpretation. If they revise the initial parse, a further question arises regarding where this revision occurs—whether it is triggered upon encountering the *tough* adjective or delayed until the verb. If the revision arises at the verb, an additional question is whether the obligatoriness of transitive verbs (e.g., *beat* requiring a direct object) can facilitate the target interpretation, as reported in previous L2 acquisition studies [3]. The current study—the first to investigate the real-time processing of the English TC—addresses these issues.

The present study tracked the eye movements of 43 L1-Korean adult learners of English as they viewed two images depicting either an agent or a theme interpretation for the matrix subject while listening to TC sentences. All participants were residing in South Korea at the time of testing and had learned English primarily in classroom settings—that is, in an English-as-a-foreign-language (EFL) context. Three conditions were manipulated, each corresponding to a different verb type: (1) optionally transitive verbs with an intransitive bias, which provide weak evidence for an object gap (k = 8); (2) obligatorily transitive verbs, which clearly cue the presence of an object gap (k = 8); and (3) prepositional verbs, in which an overt preposition signals an object gap (k = 8). After the verb, a six-word prepositional phrase followed in order to track any lingering effects after verb processing. A brief contextual sentence preceded each English TC sentence. See (1)–(3) for examples of each condition. To ensure that fixations captured linguistic interpretation rather than visual complexity, the visual scenes employed a simplified role-marking scheme: in each image, the agent was depicted in full color and the theme in silhouette (see Fig. 1). Participants also viewed 36 filler items involving sentences in either active or passive voice. To examine whether fixation patterns differed by proficiency, the 43 participants were divided into two proficiency groups (advanced, n = 23; intermediate, n = 20) based on their cloze-test scores [4].

Results showed that target-picture accuracy was 79.1% for the advanced group and 41.5% for the intermediate group. Accuracy also differed by verb type: 65.4% for intransitively-biased optionally transitive verbs, 67.7% for obligatorily transitive verbs, and 57.3% for prepositional verbs. Eye-movement data indicated that learners generally maintained an agentive interpretation from the beginning of the sentence but revised it upon hearing the verb. Learners fixated on the target picture earlier and more consistently for obligatorily transitive verbs (see Fig. 2), demonstrating sensitivity to verb-driven structural cues. In contrast, optionally transitive verbs elicited weaker shifts toward the target interpretation, and prepositional verbs elicited the lowest fixations, likely due to the reduced salience of the preposition. Proficiency further modulated these effects: advanced learners showed sustained advantages for obligatorily transitive verbs, whereas intermediate learners showed only brief verb-type differences. This pattern aligns with previous findings that higher-proficiency learners use verb subcategorization information more consistently [5].

Overall, the findings suggest that although L2 processing is initially guided by meaning-based strategies, learners—especially those with higher proficiency—can draw on verb-specific structural knowledge to revise an initial surface-based parse and engage in structure-based processing during comprehension of this complex construction [6].

(1) *Example of Condition 1: Intransitively-biased optionally transitive verbs*
(context sentence) Loopy and Pororo are wrestling near the wall.
Loopy is hard to fight behind the high and wide wall.

(2) *Example of Condition 2: Obligatorily transitive verbs*
(context sentence) Loopy and Pororo are on the trail.
Loopy is hard to catch after the big and heavy meal.

(3) *Example of Condition 3: Prepositional verbs*
(context sentence) Loopy and Pororo are in the library.
Loopy is hard to talk to in the small and quiet library.

Figure 1. *Example Visual Stimuli of Eye-tracking Experiment*

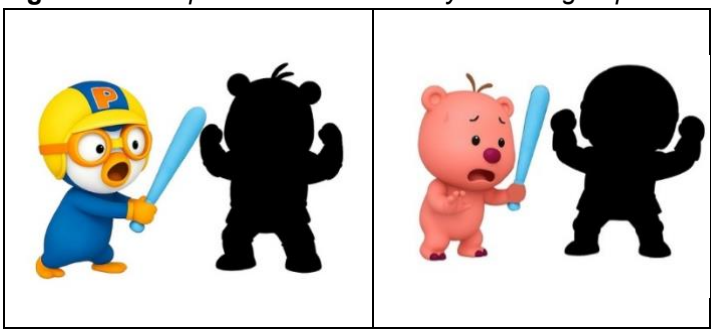
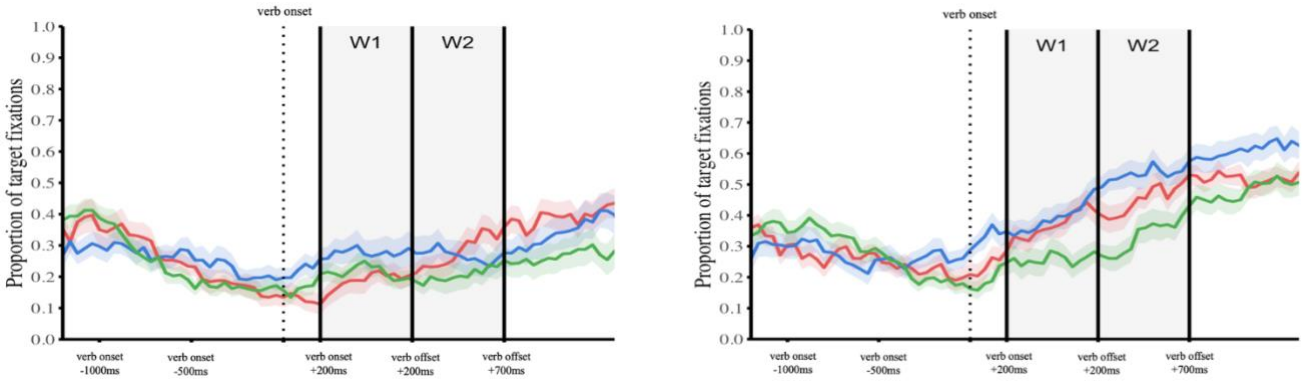


Figure 2. *Time-course Proportions of Target Fixations*



Note. Left panel: intermediate group; Right panel: advanced group. Red = intransitively-biased optionally transitive verb, blue = obligatorily transitive verb, green = prepositional verb

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